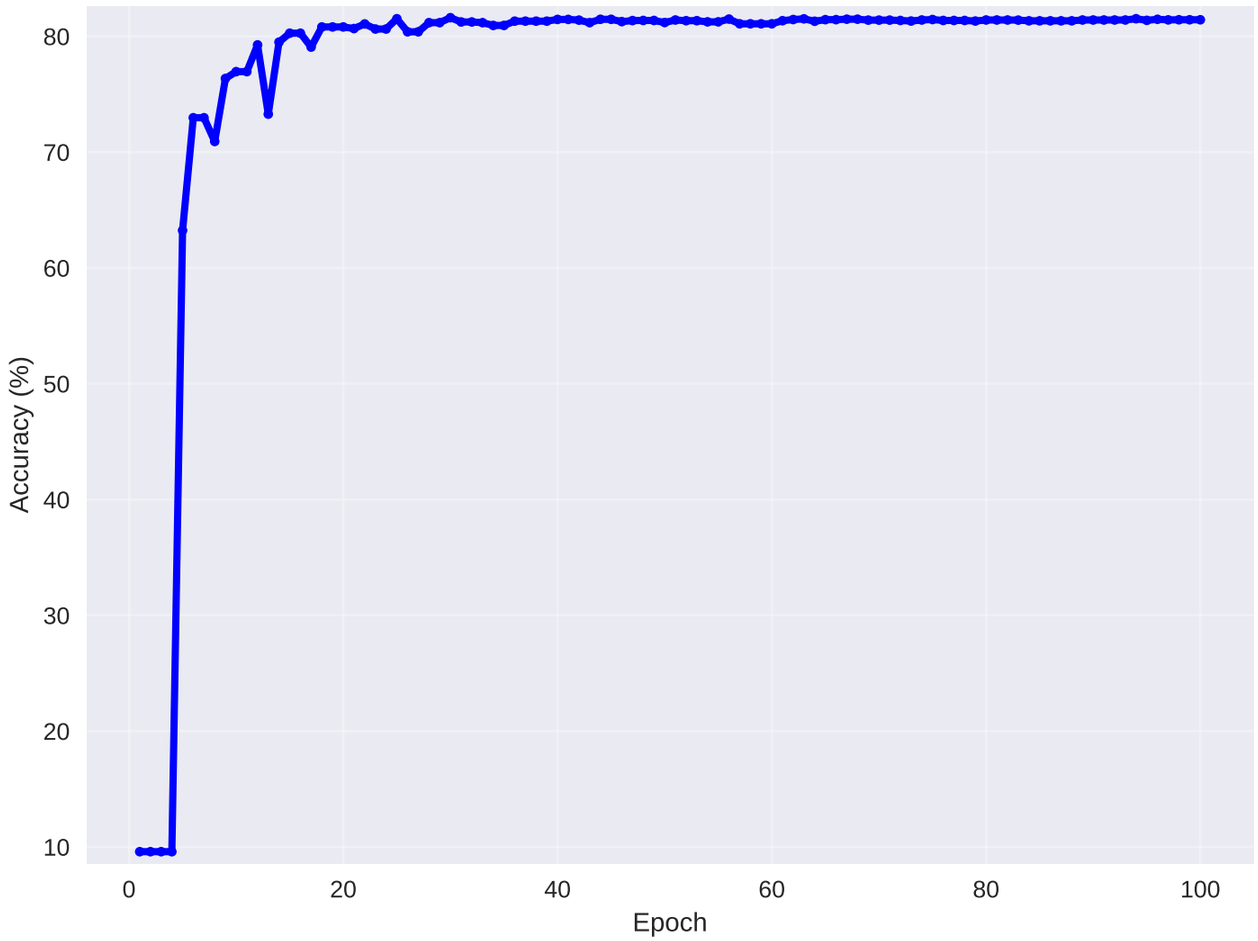
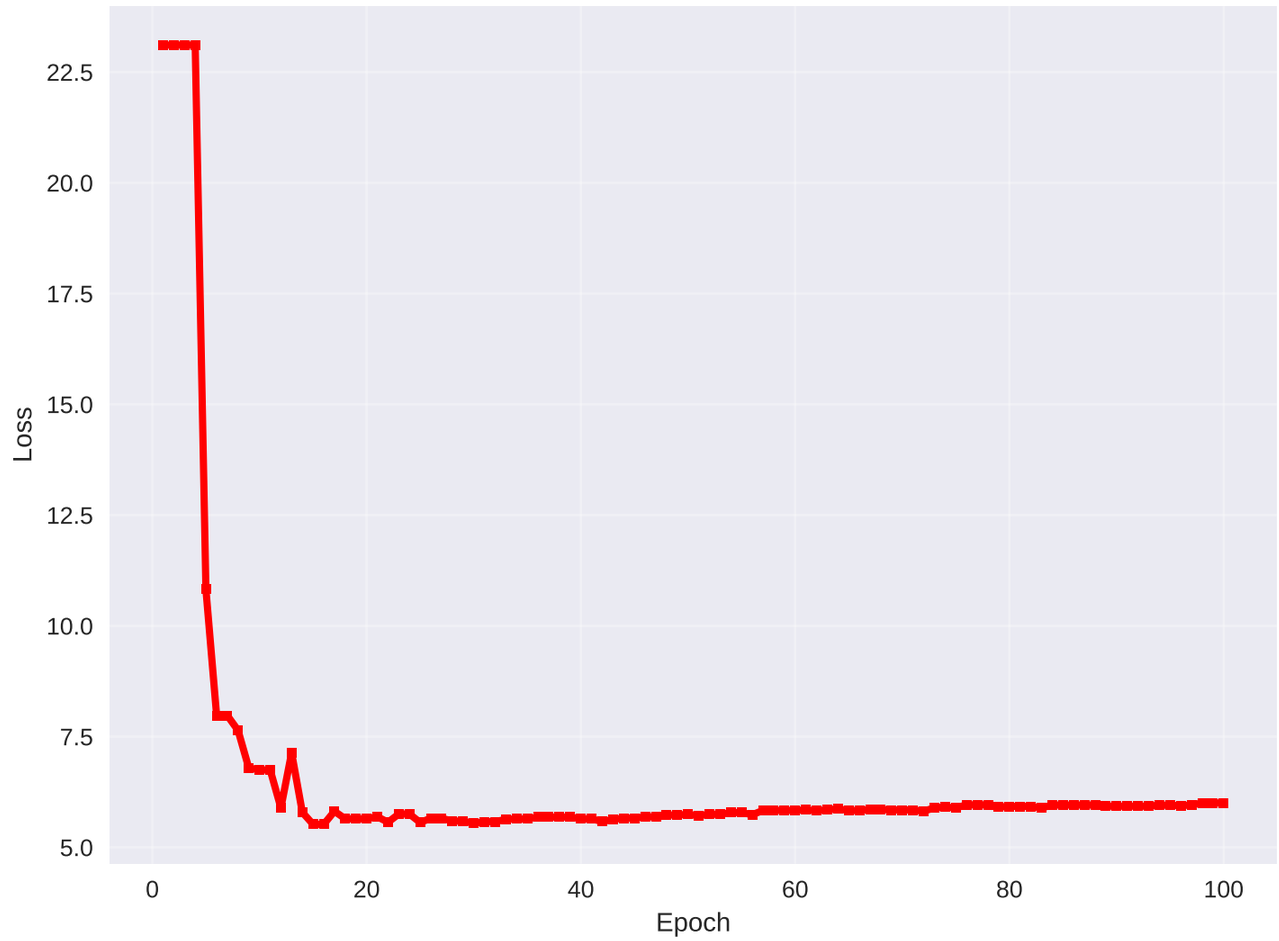
Federated Learning - Main Metrics Overview

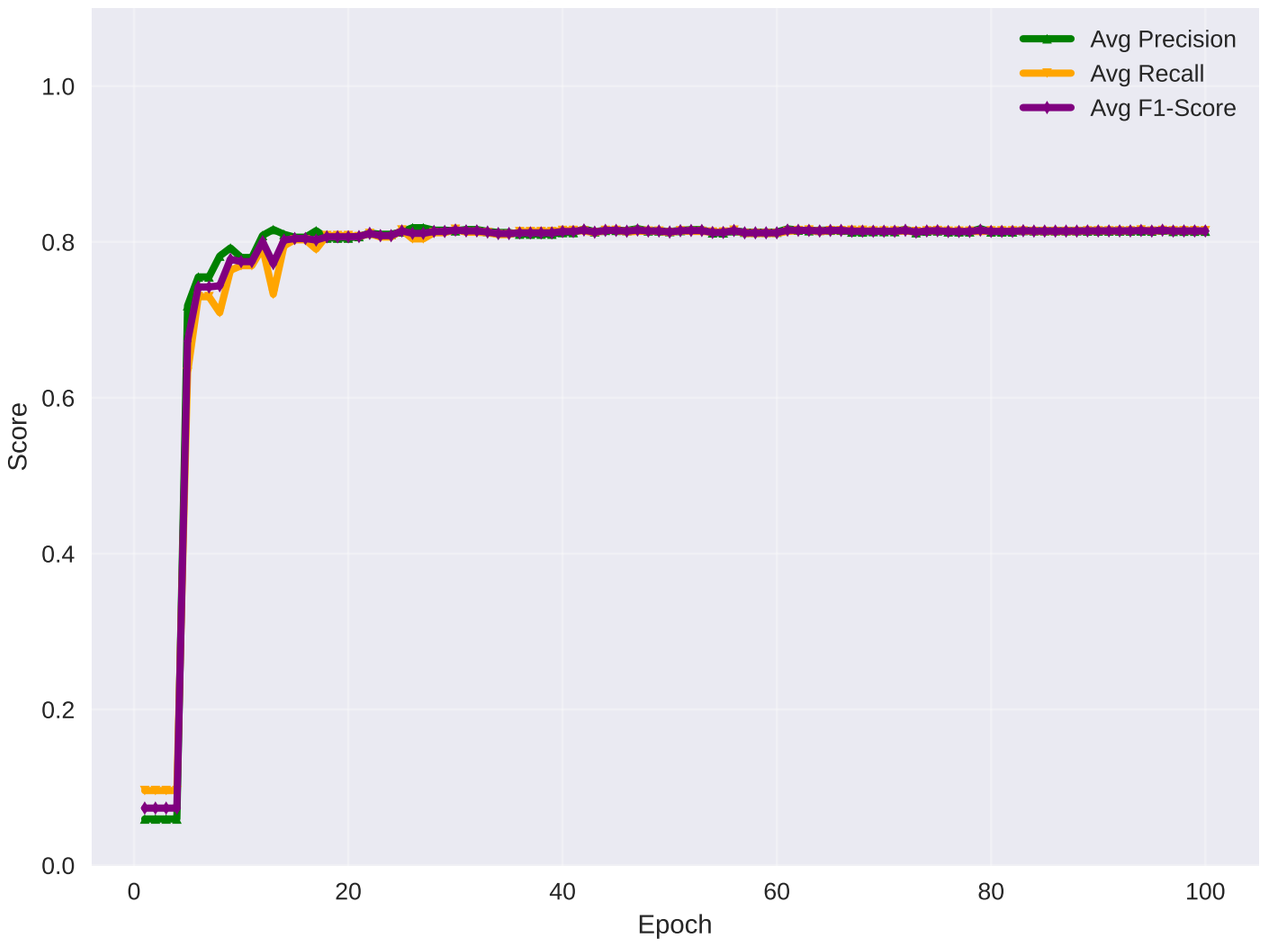
□ Global Model Accuracy Evolution



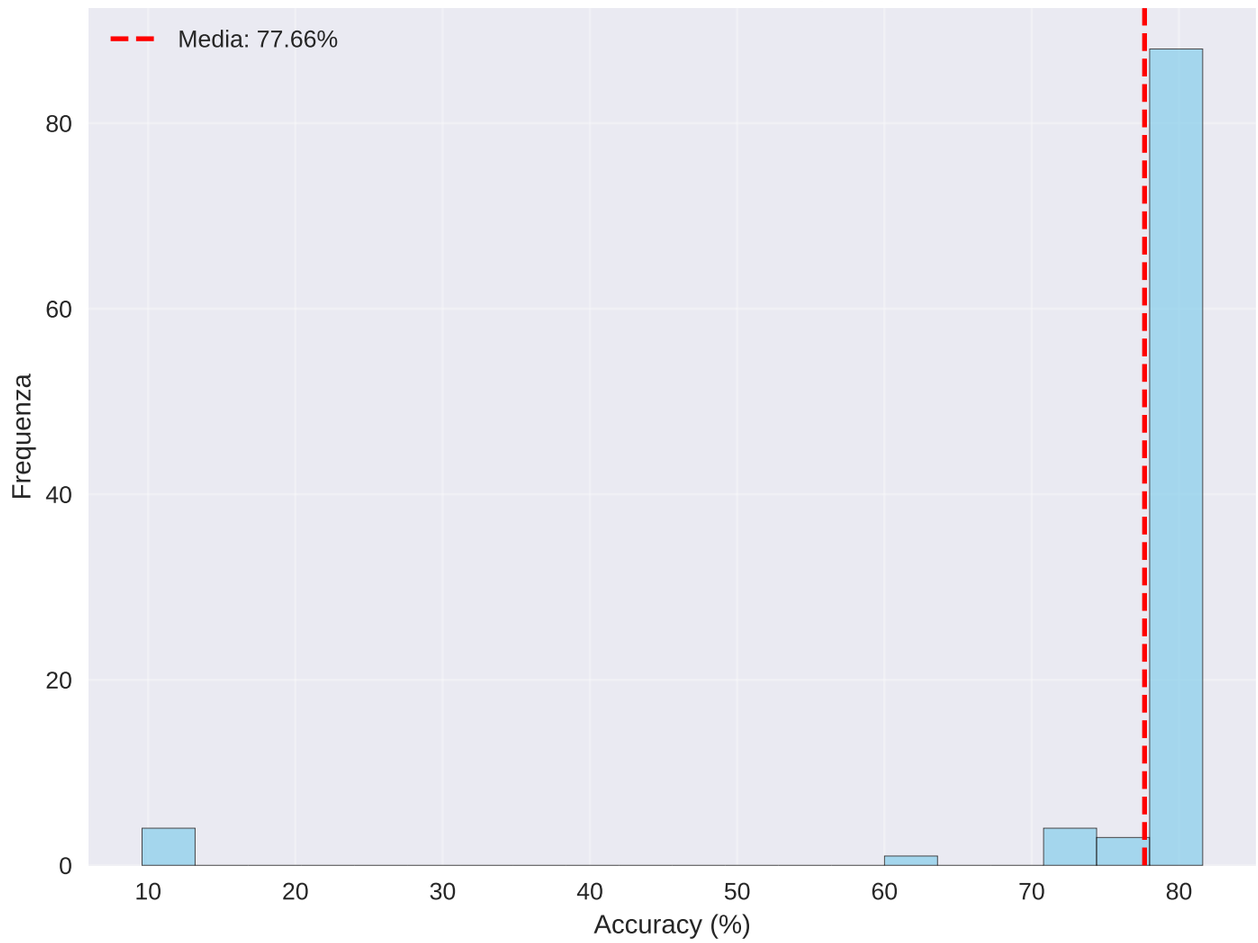
□ Global Model Loss Evolution



□ Average Precision, Recall & F1-Score

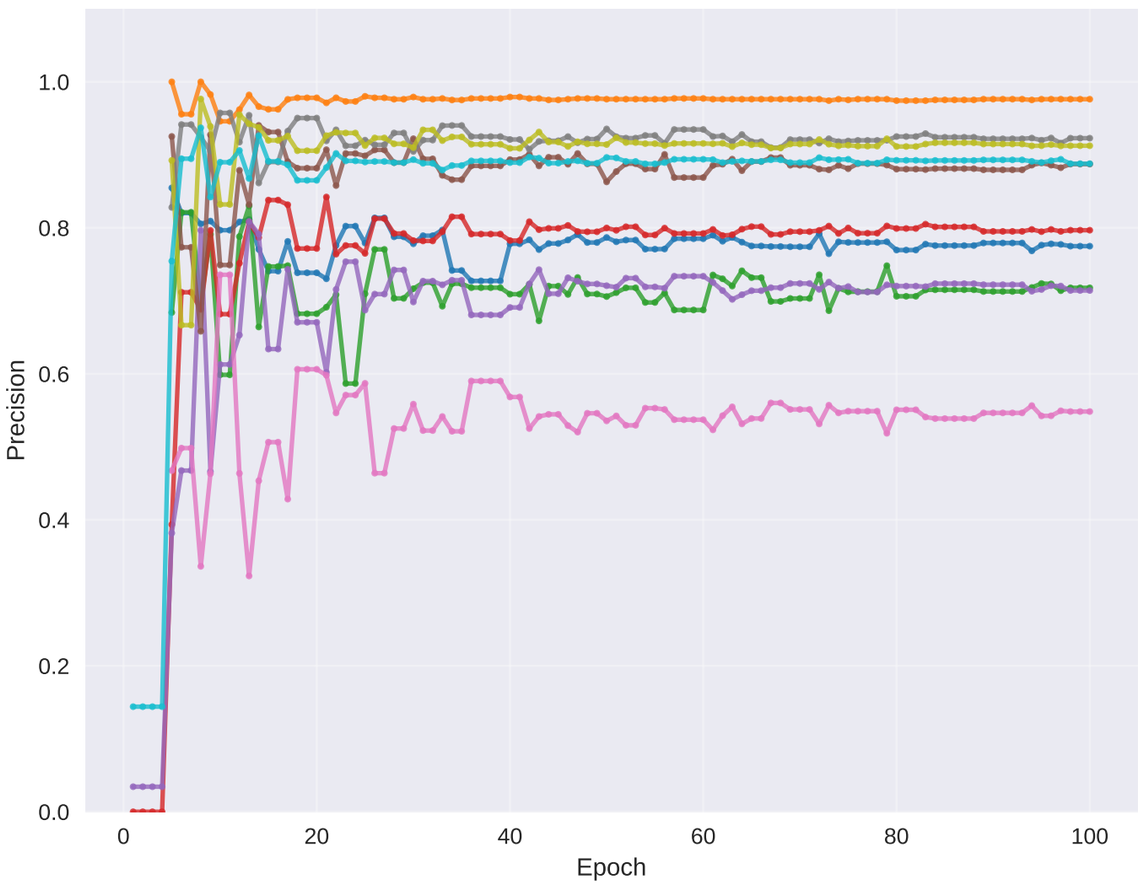


□ Distribuzione Accuracy

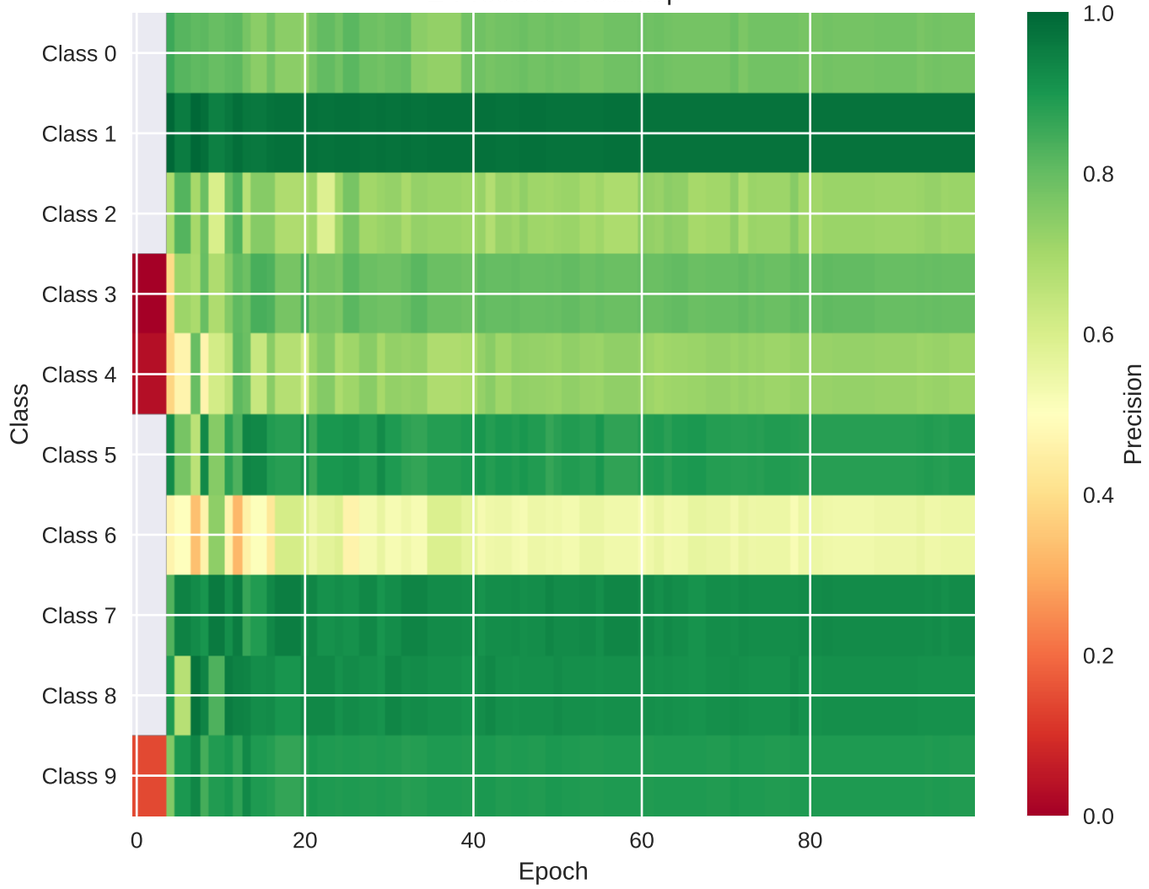


Per-Class Precision Analysis

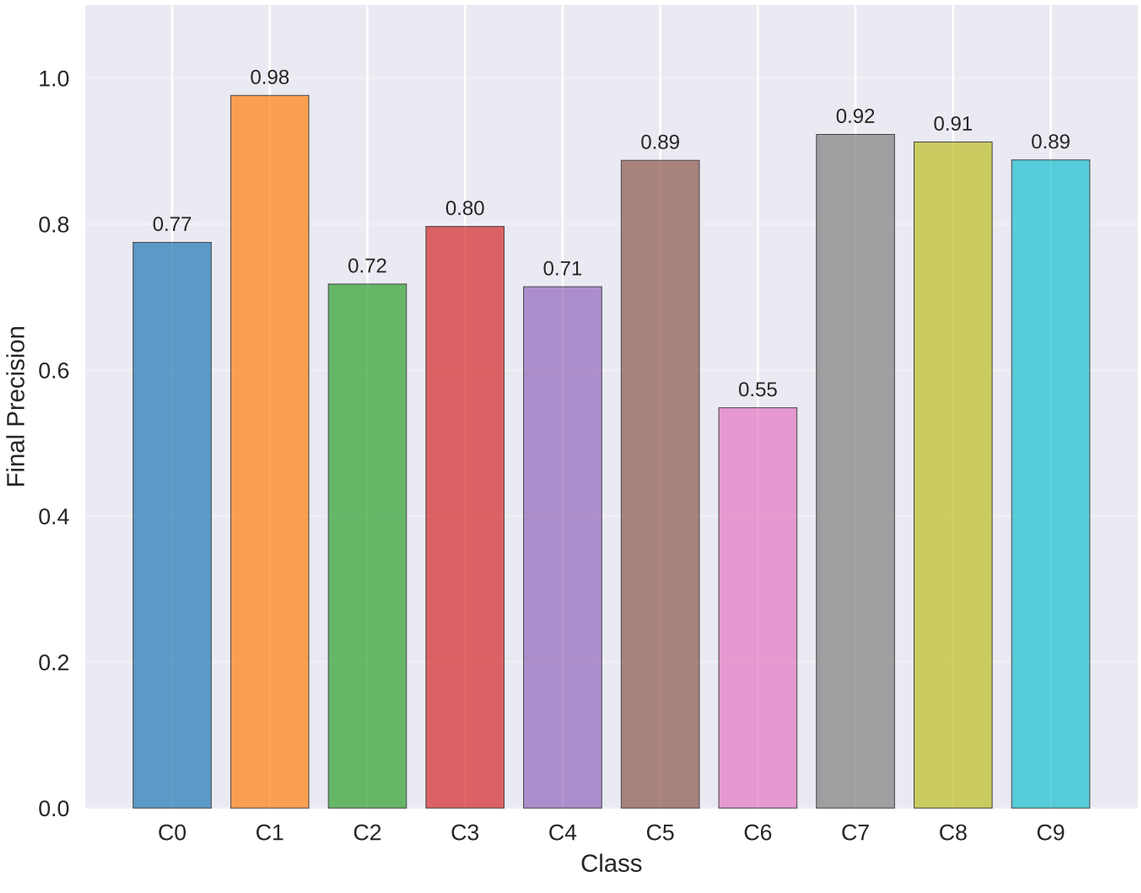
□ Per-Class Precision Evolution



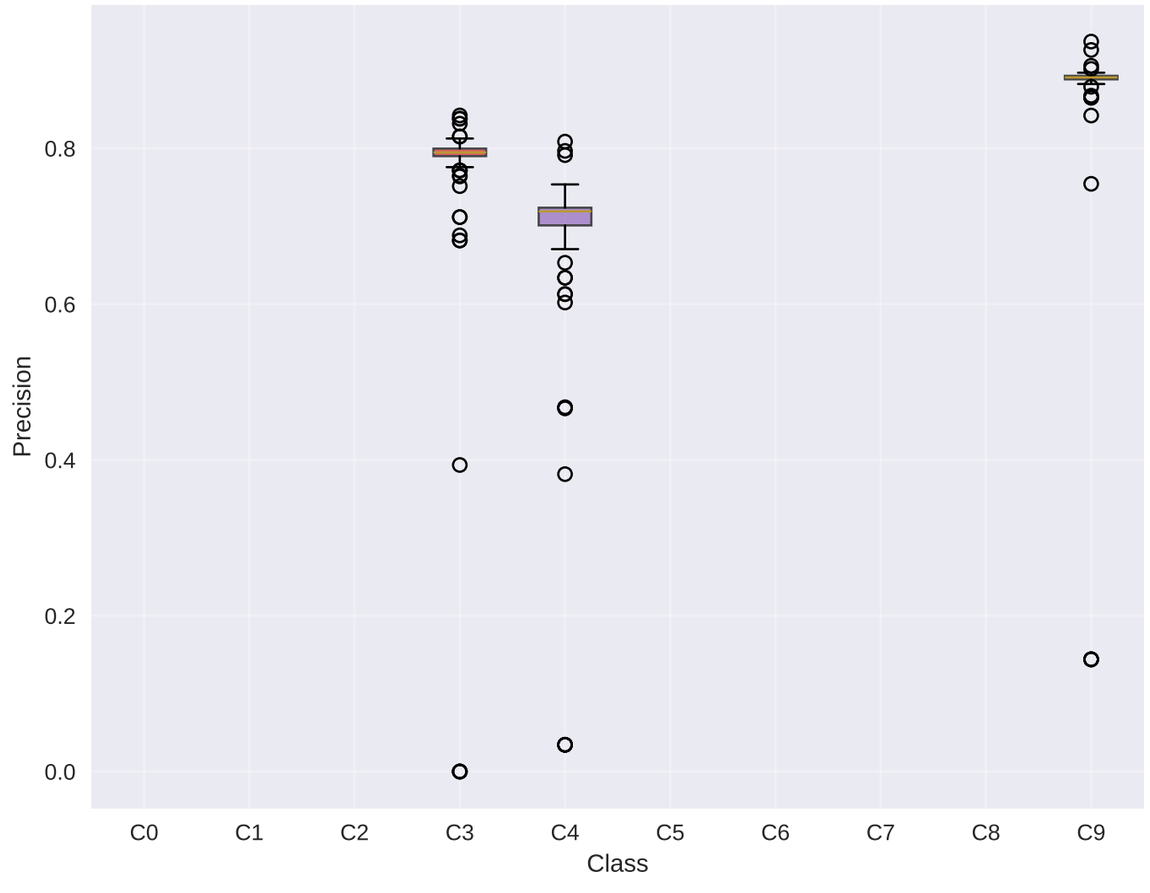
□ Precision Heatmap



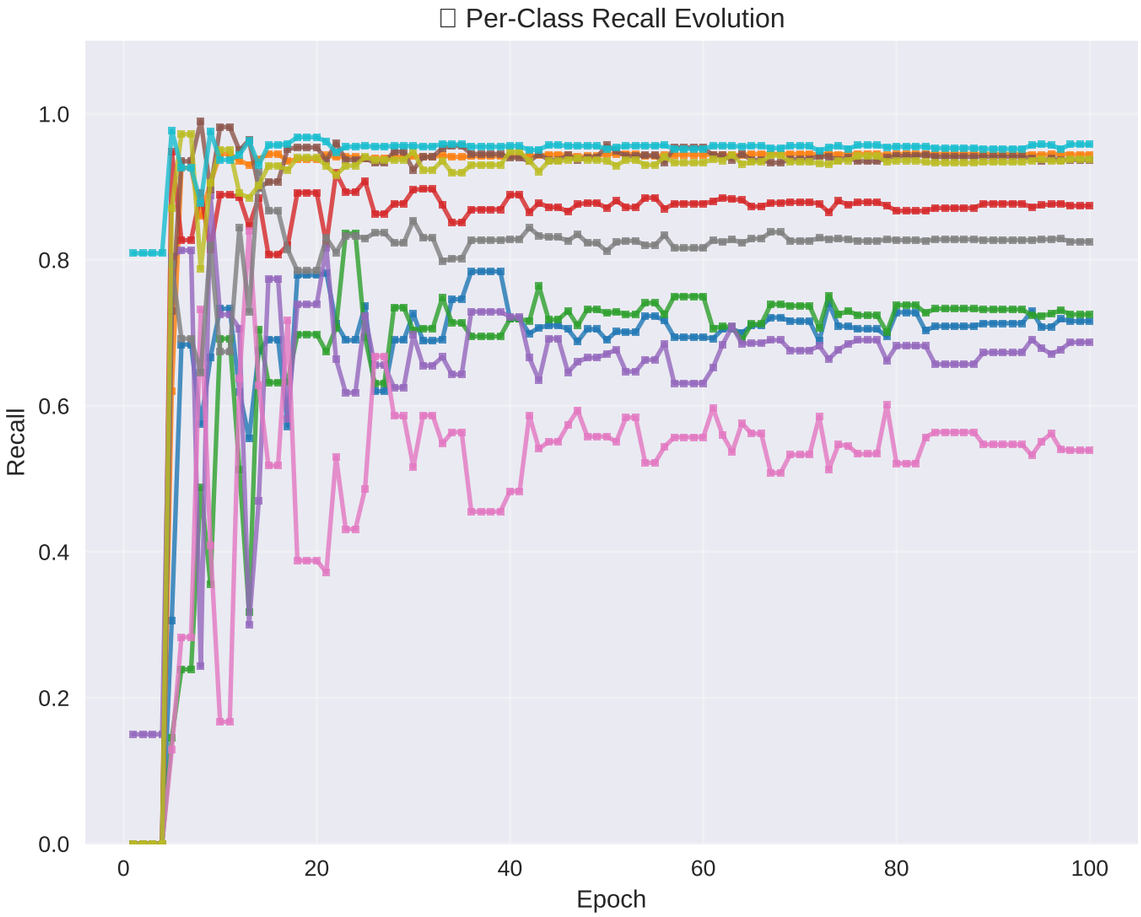
□ Final Precision per Class

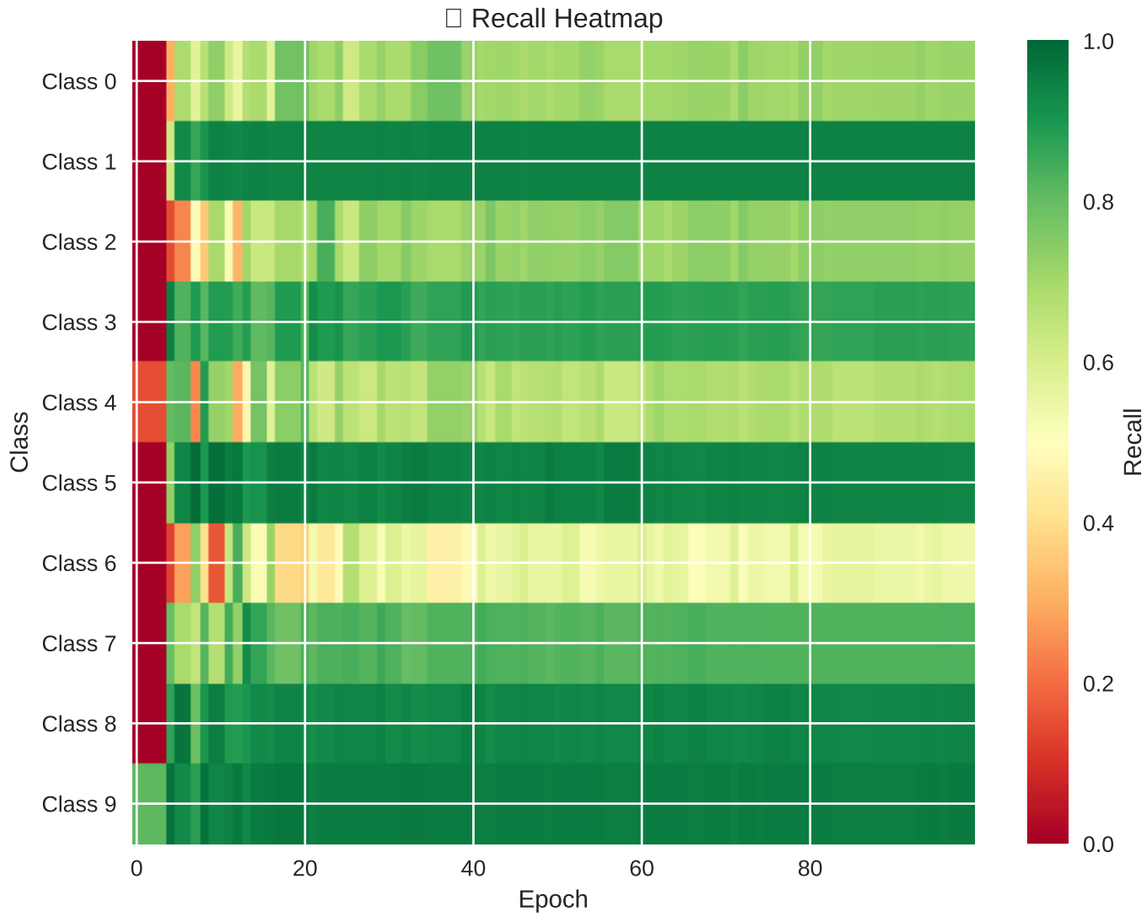


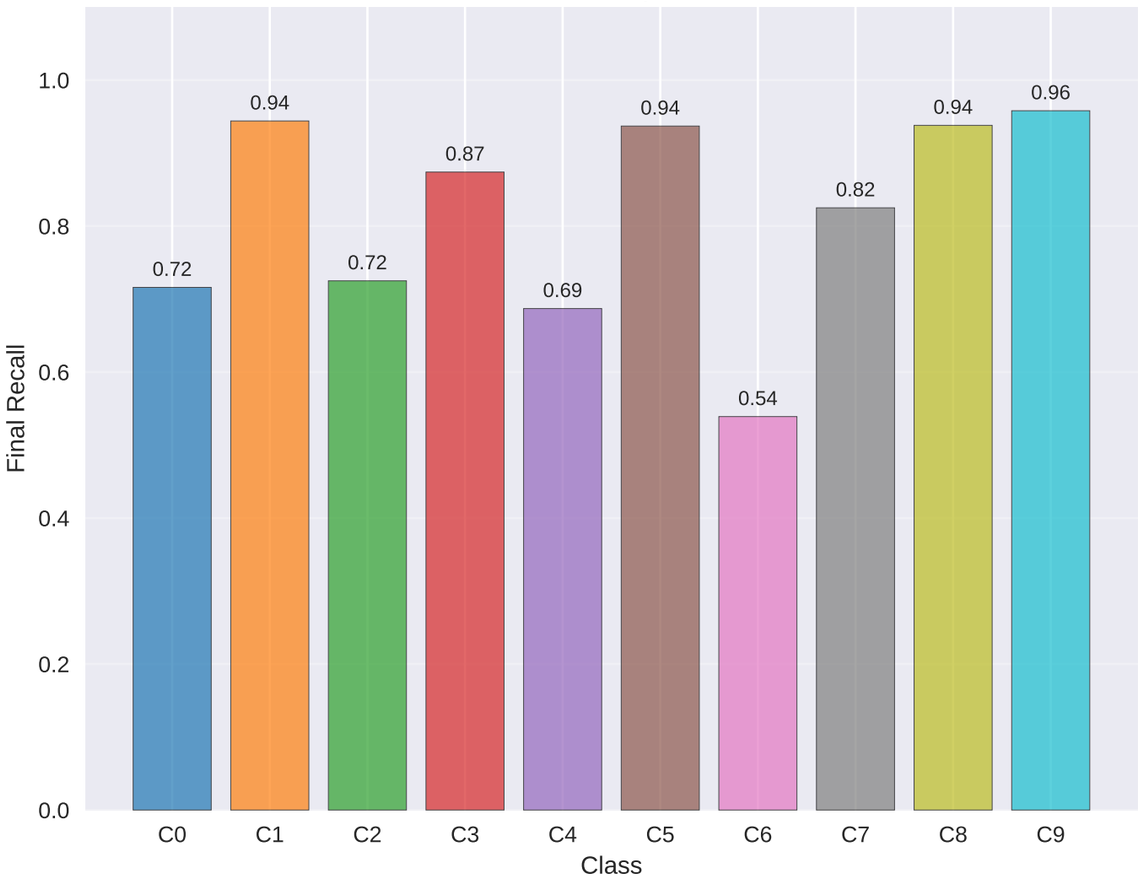
□ Precision Distribution per Class

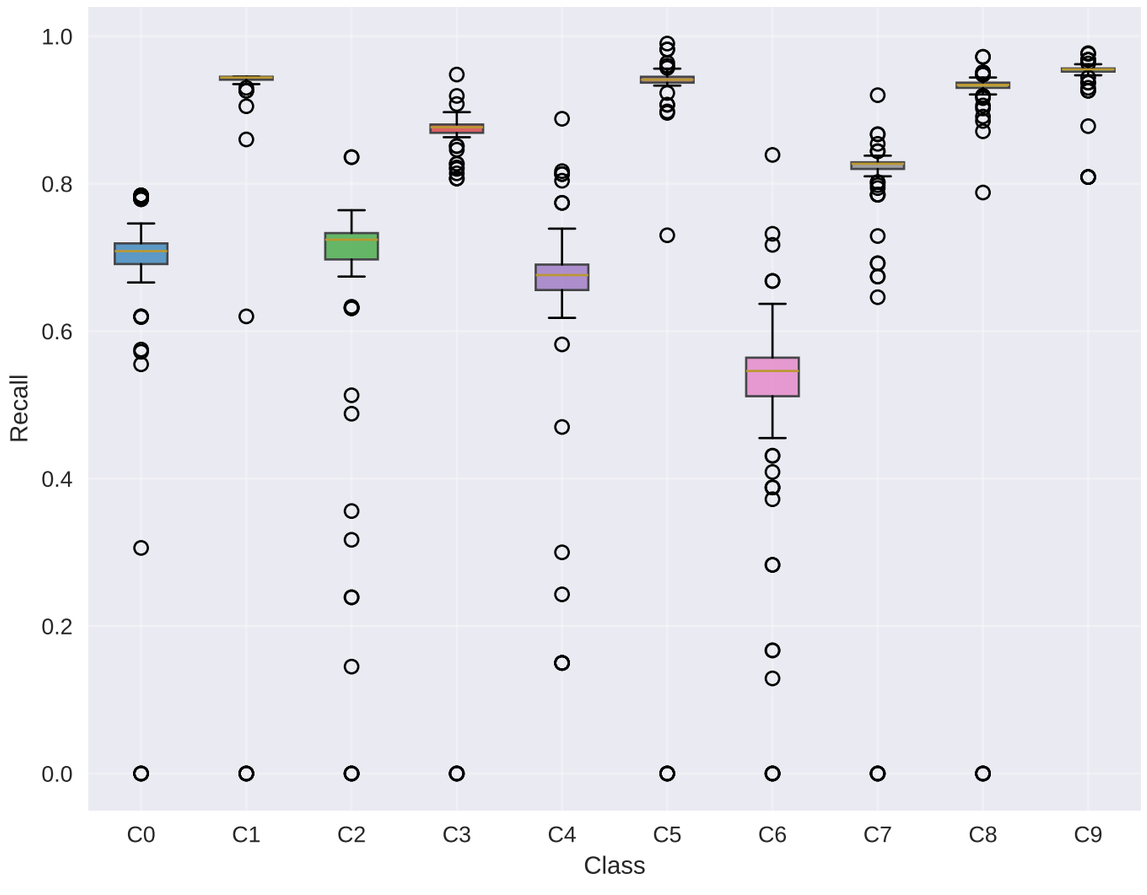


Per-Class Recall Analysis



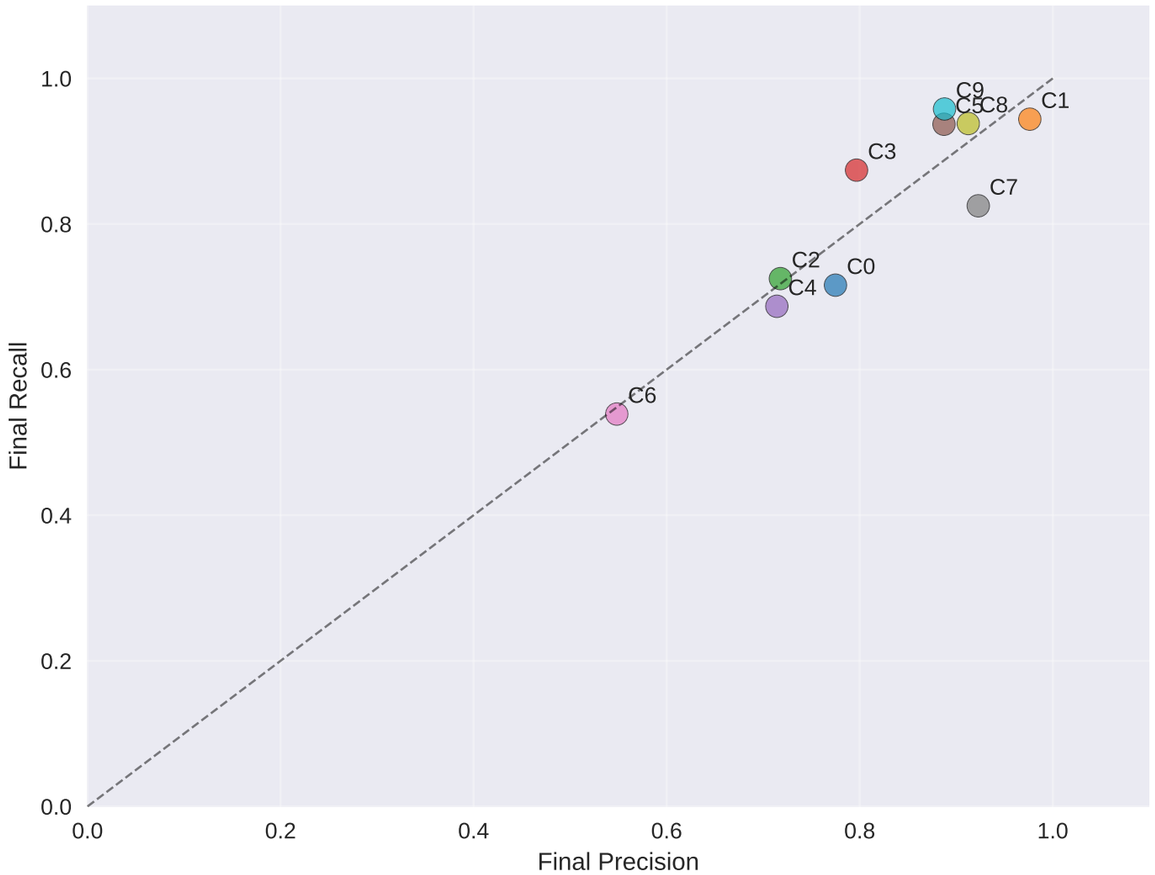




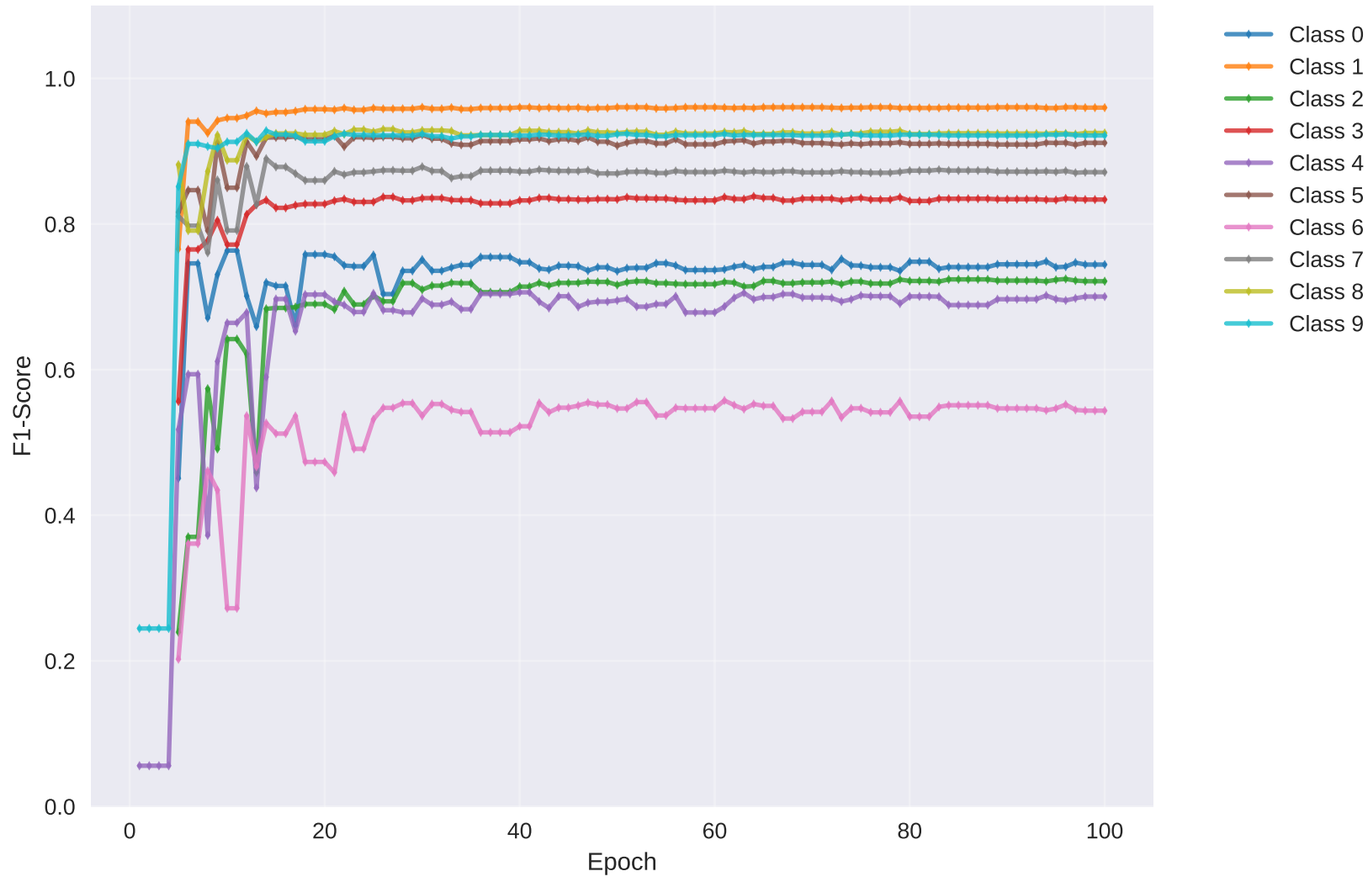


Comparative Analysis

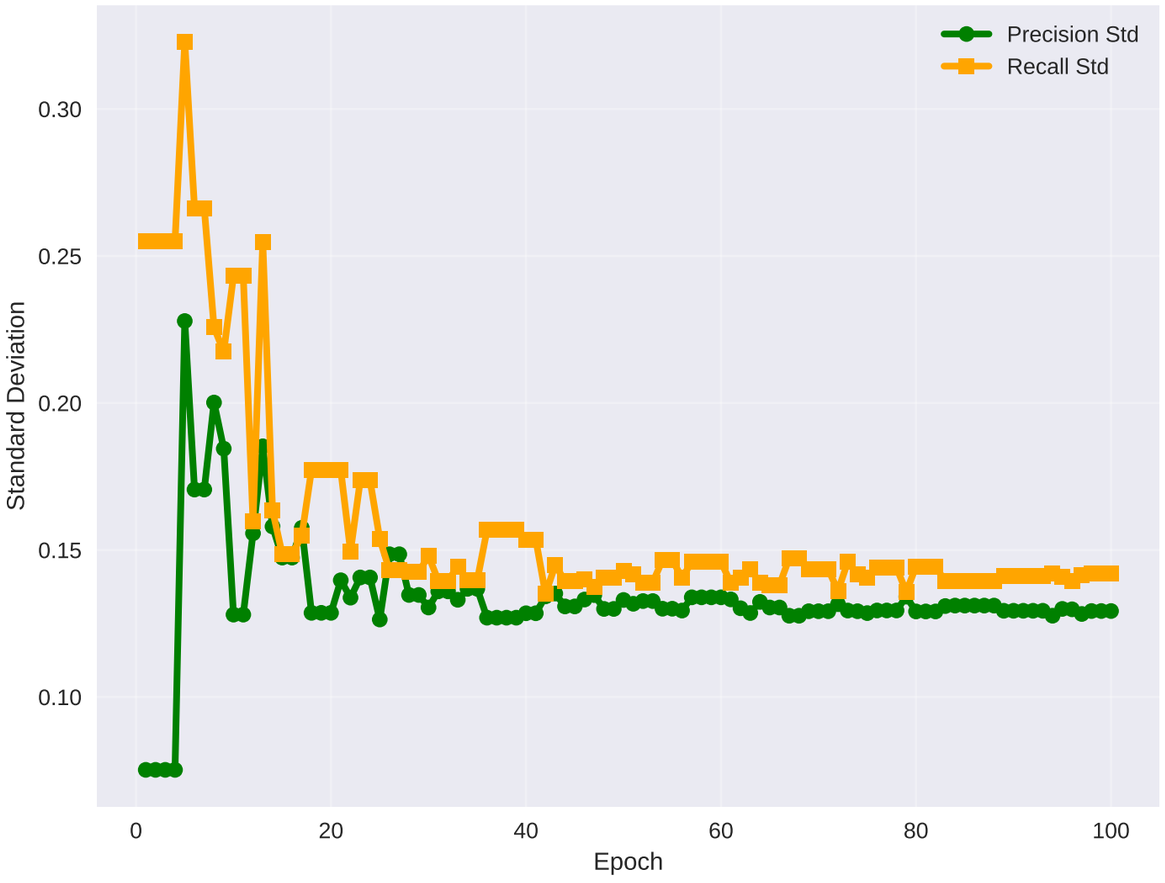
□ Precision vs Recall (Final Values)



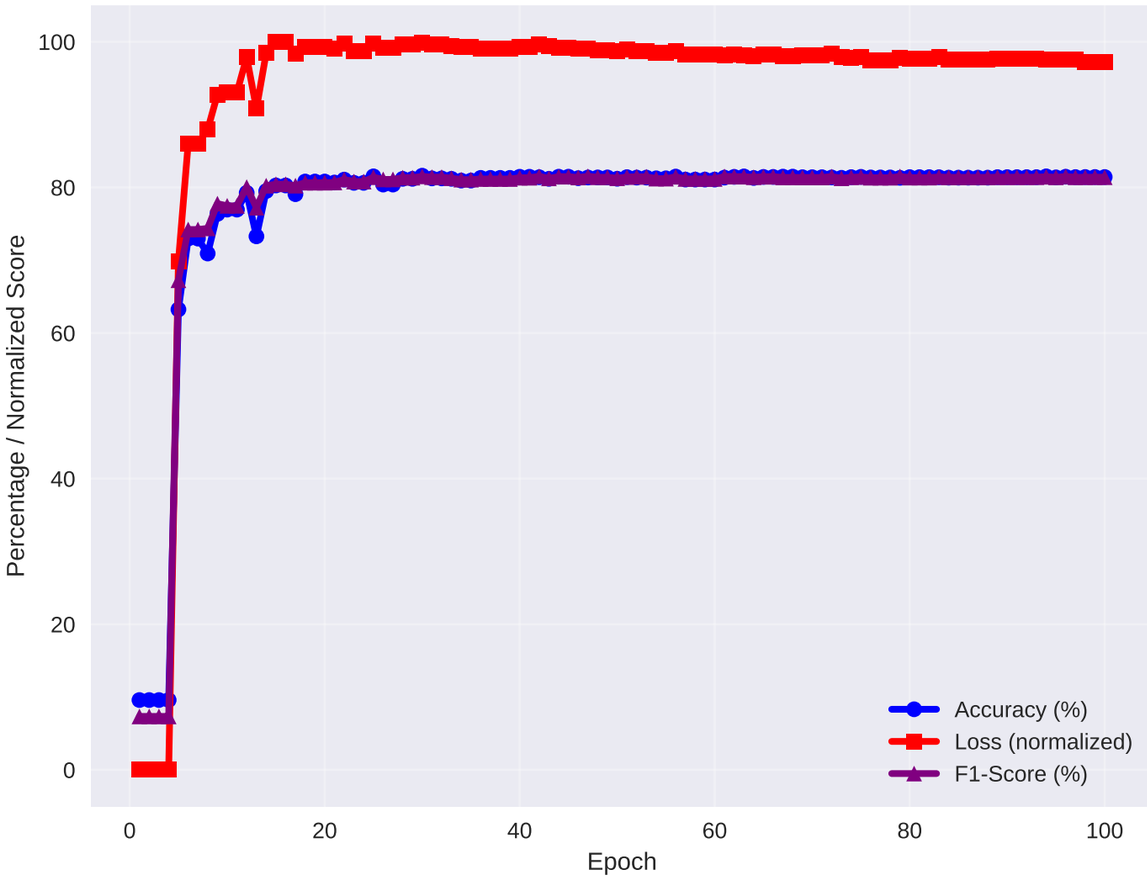
□ Per-Class F1-Score Evolution



□ Inter-Class Variance Evolution



□ Learning Curves Comparison



Performance Summary Report

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FEDERATED LEARNING - PERFORMANCE SUMMARY
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FILE ANALIZZATO: 3000_results.csv
DATA ANALISI: 2025-09-24 13:52:38
EPOCHE TOTALI: 100

METRICHE FINALI:
• Accuracy: 81.43%
• Loss: 6.0018
• Average Precision: 0.814
• Average Recall: 0.814
• Average F1-Score: 0.814

EVOLUZIONE TRAINING:
• Miglioramento Accuracy: +71.84%
• Riduzione Loss: +17.1086
• Stabilità Accuracy (CV): 0.1829

PERFORMANCE PER CLASSE:
• Migliore Precision: Classe 1 (0.976)
• Peggiore Precision: Classe 6 (0.548)
• Migliore Recall: Classe 9 (0.958)
• Peggiore Recall: Classe 6 (0.539)

ANALISI DETTAGLIATA PER CLASSE:
• Classe 0: P=0.775, R=0.716, F1=0.744
• Classe 1: P=0.976, R=0.944, F1=0.960
• Classe 2: P=0.718, R=0.725, F1=0.721
• Classe 3: P=0.797, R=0.874, F1=0.834
• Classe 4: P=0.714, R=0.687, F1=0.700
• Classe 5: P=0.887, R=0.937, F1=0.911
• Classe 6: P=0.548, R=0.539, F1=0.544
• Classe 7: P=0.923, R=0.825, F1=0.871
• Classe 8: P=0.912, R=0.938, F1=0.925
• Classe 9: P=0.888, R=0.958, F1=0.922

VALUTAZIONE QUALITATIVA:
• Performance Globale: MEDIA
• Stabilità Training: INSTABILE
• Balance Classes: SBILANCIATO

RACCOMANDAZIONI:
• Considerare aumento epoche o tuning iperparametri
• Implementare class balancing o weighted loss
• Migliorare stabilità con learning rate scheduling
```